

Problem 9.1 (MATLAB)

Non-parametric hypothesis tests

One-sample location tests

Sign test

The datafile '***dataset_problem_9_1.mat***' contains 50 iid observations from a univariate random variable, X . It is desired to do a two-sided non-parametric hypothesis test on the random variable median:

$$H_0: m = m_0 = 4$$

- Load the data
- Make visualization plots, e.g. histogram and QQ-plot
- Calculate the sample median
- Find the value of the sign test test-statistic, $S = \# \text{obs} > m_0 \sim \text{Bin}(n, \frac{1}{2})$
- Calculate the p-value for an exact (binomial) test (**binocdf**)
- Do the same using the sign test built-in command (**signtest**)
- Do an approximate test using the normal approximation to the binomial, i.e. calculate the standard normal test-statistic Z_0 and find the p-value (**normcdf**)
- Do the same using the sign test built-in command (**signtest**)
- Compare the p-values for exact and approximate tests
- Conclusion ?

Wilcoxon signed rank test

For the same dataset and null hypothesis as above, consider a somewhat more involved, but also more powerful, (Wilcoxon) signed rank test using ranks of data instead of just observing if they are below or above the null hypothesis proposed median value.

- Calculate the value of the test-statistic, W , using the recipe from the lecture notes, i.e. using signs and ranks of absolute deviations for the observations deviation from the null hypothesis proposed median value
- Using the known mean $E[W]$ and variance $V[W]$ for W , do an approximate test i.e. calculate the standard normal test-statistic Z_0 and find the p-value (**normcdf**)
- Calculate the value of the alternative test-statistic, T , using the recipe from the lecture notes
- Using the known mean $E[T]$ and variance $V[T]$ for T , do an approximate test i.e. calculate the standard normal test-statistic Z_0 and find the p-value (**normcdf**)
- Do you get the same p-value using W and T ?
- Now do the same test using the signed rank test built-in command (**signrank**)
- **Compare the p-values for the sign test** (**signtest**) **and the signed rank test** (**signrank**)
- Conclusion ?

Problem 9.2 (MATLAB)

Non-parametric hypothesis tests

Two-sample location test

Mann-Whitney U test (Wilcoxon ranksum test)

The datafile '***dataset_problem_9_2.mat***' contains iid observations from two univariate random variables, X and Y. It is desired to do a two-sided non-parametric hypothesis test on the random variable medians:

$$H_0: m_x = m_y$$

- Load the data
- Make visualization plots, e.g. histogram and QQ-plot for X and Y observations
- Calculate the sample medians
- Find the value of the sign test test-statistic, U, using the recipe from the lecture notes
- Using the known mean $E[U]$ and variance $V[U]$ for U, do an approximate test i.e. calculate the standard normal test-statistic Z_0 and find the p-value (**normcdf**)
- Calculate the value of the alternative test-statistic, W, using the recipe from the lecture notes
- Using the known mean $E[W]$ and variance $V[W]$ for W, do an approximate test i.e. calculate the standard normal test-statistic Z_0 and find the p-value (**normcdf**)
- Do you get the same p-value using U and W ?
- Now do the same test using the ranksum test built-in command (**ranksum**) and compare with above

Problem 9.3 (MATLAB)

Non-parametric hypothesis tests

Multiple sample location tests

Kruskall-Wallis test (non-parametric equivalent to ANOVA 1)

The datafile '***dataset_problem_9_3a.mat***' contains iid observations from three univariate random variables, X_1 , X_2 and X_3 . It is desired to do a two-sided hypothesis test on the random variable medians – the non-parametric equivalent to an ANOVA 1 test on random variable means.

$$H_0: m_1 = m_2 = m_3$$

- Load the data
- Make visualization plots, e.g. histogram and QQ-plot for X_1 , X_2 and X_3 observations
- Calculate the sample medians
- Find the value of the test-statistic, H , using the recipe from the lecture notes. This involves 1) sorting all observations from all 3 populations together in increasing order and give them corresponding ranks, 2) calculate average ranks of all observations in each population and 3) calculate the overall average rank for all observations
- Find the p-value for testing H_0 with the value of H and its known approximate distribution (**chi2cdf**)
- Do the same test using the Kruskal-Wallis test built-in command (**kruskalwallis**) and compare with above
- Conclusion ?

Friedman test (non-parametric equivalent to ANOVA "2")

ANOVA "2" is often called "Repeated measures ANOVA 1" or "Randomized block design ANOVA 1" and is not a true two-factor ANOVA 2 because only the column factor is tested for. The rows (often called blocks) correspond to each test object being subjected to each of the different "treatments" or "factor levels". This design gives a test of (column) factor effects with higher power than a pure ANOVA 1 test.

The datafile '***dataset_problem_9_3b.mat***' contains iid observations from four univariate random variables, X_1 , X_2 , X_3 and X_4 , $n = 50$ observations from each. The data are to be interpreted as a randomized block design as explained above, that is, similar observation indices for each random variable correspond to one test object being subjected to the four different "treatments".

It is desired to do a two-sided hypothesis test on the random variable medians:

$$H_0: m_1 = m_2 = m_3 = m_4$$

- Load the data
- Make visualization plots, e.g. histogram and QQ-plot for X_1 , X_2 , X_3 and X_4 observations
- Calculate the sample medians
- Find the value of the test-statistic, Q , using the recipe from the lecture notes. This involves 1) for each test object (block, row) sort the 4 data values in increasing order and give them corresponding ranks 1, 2, 3, and 4 (use, e.g. `tiedrank`), 2) calculate the average rank of all observations for each population, 3) calculate the overall average rank for all observations
- Find the p-value for testing H_0 with the value of Q and its known approximate distribution (`chi2cdf`)
- Do the same test using the Friedman test built-in command (`friedman`) and compare with above
- Conclusion ?

Problem 9.4 (MATLAB) **Non-parametric hypothesis tests** **Distribution tests**

One-sample Kolmogorov-Smirnov distribution test

The datafile '**dataset_problem_9_4a.mat**' contains $n = 100$ iid observations from a univariate random variable, X . It is desired to do a non-parametric hypothesis test on the plausibility of X being exponentially distributed with a specific value for the distribution parameter:

$$H_0: X \sim \text{Exp}(\lambda), \quad \lambda = \lambda_0 = 10 \Leftrightarrow \mu = \mu_0 = 1/\lambda_0 = 0.1$$

- Load the data
- Make visualization plots, e.g. 1) QQ-plot for the data against the hypothesized exponential distribution (use **makedist**), 2) histogram for the data and in the same plot the pdf for the H_0 distribution for X (use **exppdf**)
- To prepare for calculating the Kolmogorov-Smirnov test statistic, make a plot of the empirical data cdf (use **cdfplot**) together with a plot of the cdf for the H_0 distribution for X (use **expcdf**)
- Calculate the KS1 test statistic, $D = \max_x |\hat{F}(x) - F(x)|$, i.e. find (either visually or by calculation) the maximal absolute vertical difference between empirical and hypothesized cdf's
- Calculate the approximate critical value, $c(\alpha) = \sqrt{\frac{-\frac{1}{2} \log \frac{\alpha}{2}}{n}}$, where α is the chosen level of significance for the test
- Compare D with $c(\alpha)$ and conclude
- Do the same test using the Kolmogorov-Smirnov test built-in command (**kstest**) and compare results with above

Two-sample Kolmogorov-Smirnov distribution test

The datafile '**dataset_problem_9_4b.mat**' contains $n = 100$ iid observations each from two univariate random variables, $X_1 = X$ and $X_2 = Y$. It is desired to do a non-parametric hypothesis test on the plausibility of X_1 and X_2 having identical distributions and thus also having identical cdf's $F_1(x)$, $F_2(x)$

$$H_0: F_1(x) = F_2(x)$$

- Load the data
- Make visualization plots, e.g. 1) QQ-plot for the X_1 data against X_2 data), 2) histograms for the X_1 data and X_2 data

- To prepare for calculating the Kolmogorov-Smirnov test statistic, make a plot of the two empirical data cdf (use **cdfplot**) together in the same figure
- Calculate the KS2 test statistic, $D = \max_x |\hat{F}_1(x) - \hat{F}_2(x)|$, i.e. find (either visually or by calculation) the maximal absolute vertical difference between the two empirical cdf's
- Calculate the approximate critical value, $c(\alpha) = \sqrt{-\frac{1}{2} \log \frac{\alpha}{2} \left(\frac{1}{n} + \frac{1}{n} \right)}$, where α is the chosen level of significance for the test
- Compare D with $c(\alpha)$ and conclude
- Do the same test using the two-sample Kolmogorov-Smirnov test built-in command (**kstest2**) and compare results with above

Problem 9.5 (MATLAB)

Non-parametric hypothesis tests

Goodness Of Fit (GOF) test (distribution test with estimated parameters)

Goodnes Of Fit distribution test

The datafile '***dataset_problem_9_5.mat***' contains $n = 100$ iid observations from a univariate random variable, X . It is desired to do a non-parametric hypothesis test on the plausibility of X being exponentially distributed but unlike the one-sample Kolmogorov-Smirnov test now with an (unknown) estimated distribution parameter, e.g. estimated using Maximum Likelihood estimation:

$$H_0: X \sim \text{Exp}(\lambda), \quad \lambda = \hat{\lambda}_{\text{ML}} \Leftrightarrow \mu = \hat{\mu}_{\text{ML}} = \bar{x} = 1/\hat{\lambda}_{\text{ML}}$$

- Load the data
- Estimate the unknown parameter $\lambda = \hat{\lambda}_{\text{ML}} = 1/\bar{x}$. This value is then to be used for the unknown parameter in the following
- Make visualization plots, e.g. 1) QQ-plot for the data against the hypothesized exponential distribution with estimated parameter, 2) histogram for the data and in the same plot the pdf for the H_0 distribution for X (use **exppdf**), 3) plot of the empirical data cdf (use **cdfplot**) together with a plot of the cdf for the H_0 distribution for X (use **expcdf**)
- Binning: To prepare for calculating the GOF test statistic split up the domain of X (in this case $x \in [0, \infty[$), into $N_{\text{bins}} = 10$ bins. The GOF test is most reliable if the bin limits are chosen so that there is an equal number of expected observations $E_i = n/N_{\text{bins}} = 100/10 = 10$ in each bin (use **expinv** to do so).
- Now find the actual number of observations in each bin, O_i
- Calculate the test statistic value, $T = \sum_{i=1}^{N_{\text{bins}}} \frac{(O_i - E_i)^2}{E_i}$ and the degrees of freedom for T 's approximate chi-squared distribution, $df = N_{\text{bins}} - N_{\text{par}} - 1$
- Find the p-value for testing H_0 with the value of T and its known approximate distribution (**chi2cdf**)
- Do the same test using the GOF test built-in command (**chi2gof**) and compare results with above
- Conclusion ?

Problem 9.6 (MATLAB)

Non-parametric hypothesis tests

Contingency table test for categorical data

Contingency table test

The datafile '***dataset_problem_9_6.mat***' contains data for $N = 650$ citizens from a particular major city. Each individual is categorized according to two factors, A and B. Factor A corresponding to the rows of the contingency table describes the individuals highest level of education, there are 3 factor levels: high-level, middle-level or low-level education. Factor B corresponding to the columns of the contingency table describes in which city neighborhood the individual is living, there are 4 factor levels (neighborhoods).

The contingency table is containing counts of how many of the $N = 650$ citizens have a specific combination of factor A and B levels. These are the observed counts

$$O(i, j) = \# \text{ obs for category combination } (A_i, B_j)$$

It is desired to do a non-parametric hypothesis test on the plausibility of factors A and B being independent (versus education level having some impact on chosen neighborhood)

H_0 : Factors A and B are independent

- Load the data (the contingency table)
- Calculate for each factor A and B combination the expected counts, given that H_0 is true and the factors therefore can be considered independent

$$E(i, j) = \frac{\sum_{j=1}^c O(i, j) \cdot \sum_{i=1}^r O(i, j)}{\sum_{i=1}^r \sum_{j=1}^c O(i, j)} \quad \text{i.e. (row sum } i) \times (\text{column sum } j) / N$$

- Calculate the value of the test statistic, $T = \sum_{i=1}^r \sum_{j=1}^c \frac{[O(i, j) - E(i, j)]^2}{E(i, j)} \approx \chi_{(r-1)(c-1)}^2$
- Find the p-value for testing H_0 with the value of T and its known approximate distribution (**chi2cdf**)
- You may try the same test with built-in command **crosstab** but it is a little involved
- Conclusion ?